

PDF Extraction Quality Test

Validating PyMuPDF+OCR output for LLM fiscal shock detection

Fiscal Shocks Project

May 30, 2026

Overview

This notebook tests whether the local PyMuPDF PDF extraction (with OCR for scanned documents) produces text that Claude 3.5 Sonnet can effectively use for:

- **Model A:** Act detection (identify fiscal legislation passages)
- **Model B:** Motivation classification (spending-driven, counter-cyclical, etc.)
- **Model C:** Information extraction (timing, magnitudes from tables)

Setup

```
library(tidyverse)
library(here)
here::i_am("notebooks/test_text_extraction.qmd")

# Source the local extraction function
source(here("R/pull_text_local.R"))

# Load ground truth data
us_shocks <- read_csv(here("data/raw/us_shocks.csv"))
us_labels <- read_csv(here("data/raw/us_labels.csv"))
```

Phase 1: Sample Selection

We select documents with well-known fiscal acts that have clear ground truth labels.

```
# Key test cases with known acts
test_cases <- tribble(
  ~act_name, ~year, ~motivation, ~expected_url_pattern,
  "Revenue Act of 1964", 1964, "Long-run", "ERP.*1965",
  "Tax Reform Act of 1986", 1986, "Long-run",
  "ERP.*1987",
  "Economic Recovery Tax Act of 1981", 1981, "Long-
run", "ERP.*1982",
  "Tax Reduction Act of 1975", 1975, "Countercyclical",
  "ERP.*1976",
  "Revenue Act of 1950", 1950, "Spending-driven",
  "ERP.*1951"
)

test_cases
```

```
# A tibble: 5 × 4
  act_name          year
motivation expected_url_pattern
  <chr>          <dbl>
<chr>      <chr>
1 Revenue Act of 1964      1964 Long-
run      ERP.*1965
2 Tax Reform Act of 1986      1986 Long-
run      ERP.*1987
3 Economic Recovery Tax Act of 1981 1981 Long-run
ERP.*1982
4 Tax Reduction Act of 1975      1975 Countercyclical
ERP.*1976
5 Revenue Act of 1950      1950 Spending-
driven ERP.*1951
```

```
# Get actual URLs for test documents
# Using Fraser St. Louis Fed ERP archive
test_urls <- c(
  "https://fraser.stlouisfed.org/files/docs/
publications/ERP/1965/ERP_1965.pdf",
  "https://fraser.stlouisfed.org/files/docs/
publications/ERP/1982/ERP_1982.pdf"
)

cat("Test URLs:\n")
```

Test URLs:

```
cat(test_urls, sep = "\n")
```

```
https://fraser.stlouisfed.org/files/docs/
publications/ERP/1965/ERP_1965.pdf
https://fraser.stlouisfed.org/files/docs/
publications/ERP/1982/ERP_1982.pdf
```

Phase 2: Extract Sample Documents

```
# Extract text from sample PDFs using local PyMuPDF+OCR
# This may take 5-15 minutes for scanned documents
# (OCR needed)
message("Starting local extraction...")

sample_results <- pull_text_local(
  pdf_url = test_urls,
  output_dir = here("data/extracted"),
  workers = 4,
  ocr_dpi = 200
)

message("Extraction complete!")
```

Extraction Summary

```
sample_results |>
  mutate(
    doc_name = basename(test_urls)
  ) |>
  select(doc_name, n_pages, ocr_used, extraction_time,
         extracted_at)
```

```
# A tibble: 2 × 5
  doc_name      n_pages ocr_used extraction_time
extracted_at
  <chr>        <int> <lgl>          <dbl>
<dtm>
1 ERP_1965.pdf    300 TRUE          277.
2026-01-17 15:35:36
```

```
2 ERP_1982.pdf      368 TRUE      368.
2026-01-17 15:35:36
```

Phase 3: Text Quality Inspection

3.1 Basic Text Readability

```
# Show sample text from first document
cat("=== Sample text from ERP 1965 (first 2000 chars)
===\n\n")
```

```
=== Sample text from ERP 1965 (first 2000 chars) ===
```

```
first_doc_text <- unlist(sample_results$text[[1]])
cat(substr(paste(first_doc_text, collapse = "\n\n"),
1, 2000))
```

```
_ Jaturday 1960
|
Together With
THE ANNUAL REPORT
of the
COUNCIL OF ECONOMIC ADVISERS
Digitized for FRASER
http://fraser.stlouisfed.org,
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Transmitted to the Congress
January 1965
TOGETHER WITH
THE ANNUAL REPORT
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UNITED STATES GOVERNMENT PRINTING OFFICE
WASHINGTON: 1965
Digitized for FRASER
<http://fraser.stlouisfed.org/>
Federal Reserve Bank of St. Louis

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<http://fraser.stlouisfed.org/>
Federal Reserve Bank of St. Louis

CONTENTS	
Economic REPORT OF THE PRESIDENT	
Page	
Proccess TowarD Our Economic GOALs.....--	.000-5
3	
Full Employment... 0.0.0.0... 0-0.	
o cence eee eee	
3	
Rapid Growth. 00.00.	
e eens	
4	
Price Stability... 22.60. .eccccccc	
4	
Balance	of

Payments Equilibrium.065
 4
 Consistency of Our Goals... 2.0.0.0... 20
 e eee eee eee
 5
 Tue ROLE oF Economic POLicy.....0.
 000 cece eee eee
 5
 THE UNFINISHED TASKS... 2.00.00
 o
 eee
 eee eee ee
 7
 Economic PROSPECTS FOR 1965.....
 0000 cece eee eee eee
 9
 Federal Fiscal Policy... 2... 0.00.
 c cece eee
 9
 Progress Toward
 Full Employment.....05.
 10
 CoMBATING RECESSIONS...
00.000000 cece eect
 eee
 10
 Monetary POLicy IN 1965.....
 000.0 c cece eee
 ees
 11
 MaInTAINING WAGE-PRICE STABILITY...
 0.2.0...000.00 eeeee
 12
 INTERNATIONAL ECONOMIC POLICIES.0...000
 0000 e eee eee
 13
 Restoring Balance in Our External
 Payments.....
 13
 Building a Stronger World
 Order.....0000005
 14
 Manpower POLiIciEs FOR A FLEXIBLE

```

ECONOMY.....-...-
15
U.S. Employment Service... 0... ..000. 0.0
15
Manpower Training..... 0000s cece
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15
Private Pension and Welfare
Funds.....0-.2005
15
MaiInTAINING INCOMES OF THE
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```

3.2 Check for Known Act Names

```

# Acts we expect to find in these documents
expected_acts <- c(
  "Revenue Act of 1964",
  "Economic Recovery Tax Act",
  "Tax Equity and Fiscal Responsibility Act"
)

# Search for act names in extracted text
all_text <- paste(unlist(sample_results$text),
collapse = " ")

act_detection <- map_dfr(expected_acts, function(act)
{
  found <- str_detect(all_text, fixed(act))
  tibble(
    act_name = act,
    found = found,
    mentions = sum(str_count(all_text, fixed(act)))
  )
})

act_detection

```

```

# A tibble: 3 × 3
  act_name found mentions
  <chr>    <lgl>    <int>
1 Revenue Act of 1964 TRUE      4

```

2 Economic Recovery Tax Act	TRUE	17
3 Tax Equity and Fiscal Responsibility Act	FALSE	0

```
# Act detection rate
detection_rate <- mean(act_detection$found)
cat(sprintf("Act detection rate: %.0f%% (%d/%d)\n",
            detection_rate * 100,
            sum(act_detection$found),
            nrow(act_detection)))
```

Act detection rate: 67% (2/3)

```
if (detection_rate < 0.8) {
  warning("Act detection rate below 80% - may need to
  check extraction quality")
}
```

3.3 Numeric Value Preservation

```
# Check for dollar amounts (critical for Model C)
dollar_pattern <- "\\$\\s*\\d+\\.?\\d*\\s*(billion|
million|B|M)"
year_pattern <- "\\b(19[4-9]\\d|20[0-2]\\d)\\b"

numeric_metrics <- tibble(
  metric = c("Dollar amounts", "Year mentions",
"Percentage values"),
  pattern = c(dollar_pattern, year_pattern, "\\d+\\.?\\d*\\s*percent"),
  count = c(
    sum(str_count(all_text, regex(dollar_pattern,
ignore_case = TRUE))),
    sum(str_count(all_text, year_pattern)),
    sum(str_count(all_text, regex("\\d+\\.?\\d*\\s*percent", ignore_case = TRUE)))
  )
)

numeric_metrics
```



```
# A tibble: 3 × 3
  metric pattern
count
  <chr>      <chr>      <int>
1 Dollar amounts "\\$\\s*\\d+\\.?\\d*\\s*(billion|
million|B|M)" 214
2 Year mentions "\\b(19[4-9]\\d|20[0-2]\\d)\\b"
6201
3 Percentage values "\\d+\\.?\\d*\\s*percent"
527
```

3.4 Fiscal Policy Terms

```
# Check for key fiscal policy terminology
fiscal_terms <- c(
  "tax cut", "tax reduction", "fiscal policy",
  "federal budget", "deficit", "expenditure",
  "revenue", "appropriation"
)

term_counts <- map_dfr(fiscal_terms, function(term) {
  tibble(
    term = term,
    count = sum(str_count(all_text, regex(term,
ignore_case = TRUE)))
  )
})

term_counts |>
  filter(count > 0) |>
  arrange(desc(count))
```

```
# A tibble: 8 × 2
  term count
  <chr>   <int>
1 expenditure 246
2 deficit 192
3 revenue 74
4 tax cut 50
5 federal budget 32
6 fiscal policy 31
7 tax reduction 17
8 appropriation 4
```

Phase 4: LLM Readiness Tests

4.1 Passage Length Analysis

```
# Check if pages are reasonable length for LLM context
first_doc_pages <- sample_results$text[[1]]
page_lengths <- map_int(first_doc_pages, nchar)

tibble(
  metric = c("Min page length", "Max page length", "Mean
page length", "Total chars"),
  value = c(min(page_lengths), max(page_lengths),
mean(page_lengths), sum(page_lengths))
) |>
  mutate(value = scales::comma(value))
```

```
# A tibble: 4 × 2
  metric      value
  <chr>      <chr>
1 Min page length 85
2 Max page length 7,831
3 Mean page length 3,158
4 Total chars    947,522
```

4.2 Token Estimation

```
# Rough token estimate (1 token ~ 4 chars for English)
total_chars <-
sum(map_int(unlist(sample_results$text), nchar))
estimated_tokens <- total_chars / 4

cat(sprintf("Estimated      tokens:      %s\n",
scales::comma(estimated_tokens)))
```

```
Estimated tokens: 520,314
```

```
cat(sprintf("Claude   3.5   Sonnet   context:   200K
tokens\n"))
```

```
Claude 3.5 Sonnet context: 200K tokens
```

```
cat(sprintf("Fits in context: %s\n",
           ifelse(estimated_tokens < 200000, "YES",
                  "NO - need chunking")))
```

```
Fits in context: NO - need chunking
```

4.3 Sample Passage for Model A Test

```
# Find a passage mentioning a fiscal act
first_doc_pages <- sample_results$text[[1]]

act_passages <- map_dfr(seq_along(first_doc_pages),
function(i) {
  page_text <- first_doc_pages[[i]]
  if (str_detect(page_text, regex("act of \\d{4}",
ignore_case = TRUE))) {
    tibble(
      page = i,
      text = str_trunc(page_text, 500),
      has_act_mention = TRUE
    )
  } else {
    NULL
  }
})

if (nrow(act_passages) > 0) {
  cat("=== Sample passage with act mention (for Model
A) ===\n\n")
  cat(act_passages$text[1])
} else {
  cat("No passages with 'Act of YYYY' pattern found")
}
```

```
=== Sample passage with act mention (for Model A) ===
```

```
or a lack of confidence in the dollar.
Since 1946, therefore, we have
come to recognize that the mandate of the Employment
Act implies a
series of objectives closely related to the goal of
full employment:
```

```

-rapid growth,
- price stability, and
-equilibrium in our balance of payments.
Rapp GRowTH
True prosperity means more than the full use of the
productive
powers available at any given time.
It also means the rapid expansion
of those powers.
In the long run, it is only a growth of over-all ...

```

Phase 5: Quality Metrics Summary

```

# Compile all quality metrics
total_pages <- sum(sample_results$n_pages)
total_tables <- 0 # PyMuPDF doesn't extract tables
separately

# Convert all values to character to avoid type mismatch
in tribble
quality_report <- tribble(
  ~metric, ~value, ~target, ~status,
    "Documents", "extracted",
as.character(nrow(sample_results)), "2", "PASS",
  "Pages extracted", as.character(total_pages), ">0",
    ifelse(total_pages > 0, "PASS", "FAIL"),
    "OCR
used", as.character(sum(sample_results$ocr_used)),
"as needed", "INFO",
  "Act name recall", sprintf("%.0f%%", detection_rate
* 100), ">80%",
    ifelse(detection_rate >= 0.8, "PASS", "FAIL"),
    "Dollar amounts found",
as.character(numeric_metrics$count[1]), ">0",
    ifelse(numeric_metrics$count[1] > 0, "PASS",
"WARN"),
    "Year mentions",
as.character(numeric_metrics$count[2]), ">10",
    ifelse(numeric_metrics$count[2] > 10, "PASS",
"WARN"),
  "Fits in LLM context", ifelse(estimated_tokens <
200000, "YES", "NO"), "YES",
    ifelse(estimated_tokens < 200000, "PASS", "WARN")

```

```
)

quality_report |>
  gt::gt()
```

metric	value	target	status
Documents extracted	2	2	PASS
Pages extracted	668	>0	PASS
OCR used	2	as needed	INFO
Act name recall	67%	>80%	FAIL
Dollar amounts found	214	>0	PASS
Year mentions	6201	>10	PASS
Fits in LLM context	NO	YES	WARN

Conclusions

```
pass_count <- sum(quality_report$status == "PASS")
warn_count <- sum(quality_report$status == "WARN")
fail_count <- sum(quality_report$status == "FAIL")
info_count <- sum(quality_report$status == "INFO")

cat(sprintf("\n=== QUALITY ASSESSMENT ===\n"))
```

```
=== QUALITY ASSESSMENT ===
```

```
cat(sprintf("PASS: %d | WARN: %d | FAIL: %d | INFO:
%d\n\n",
           pass_count, warn_count, fail_count,
           info_count))
```

```
PASS: 4 | WARN: 1 | FAIL: 1 | INFO: 1
```

```
if (fail_count == 0) {
  cat("Extraction quality is SUFFICIENT for LLM
processing\n")
  cat("      Proceed with full pipeline:
```

```
tar_make(us_text)\n")
} else {
  cat("Extraction quality needs IMPROVEMENT before full
run\n")
  cat("  Review failed metrics and adjust extraction
settings\n")
}
```

```
Extraction quality needs IMPROVEMENT before full run
  Review failed metrics and adjust extraction settings
```

Next Steps

Based on the quality assessment:

1. **If PASS:** Run `tar_make(us_text)` to extract all 350 documents
2. **If WARN:** Review warnings but proceed with caution
3. **If FAIL:** Debug extraction issues before full run

Recommended Actions

```
if (fail_count > 0) {
  cat("Issues to address:\n")
  quality_report |>
    filter(status == "FAIL") |>
    pull(metric) %>%
    paste("-", .) |>
    cat(sep = "\n")
}
```

```
Issues to address:
- Act name recall
```

Extraction Performance

```
# Summary of extraction times
cat("=== Extraction Performance ===\n")
```

```
=== Extraction Performance ===
```

```
cat(sprintf("Total          documents:          %d\n",
nrow(sample_results)))
```

Total documents: 2

```
cat(sprintf("Total          pages:
%d\n", sum(sample_results$n_pages)))
```

Total pages: 668

```
cat(sprintf("Total extraction time: %.1f seconds\n",
            sum(sample_results$extraction_time, na.rm
= TRUE)))
```

Total extraction time: 644.2 seconds

```
cat(sprintf("Average   time   per   document:   %.1f
seconds\n",
            mean(sample_results$extraction_time, na.rm
= TRUE)))
```

Average time per document: 322.1 seconds

```
cat(sprintf("Average time per page: %.2f seconds\n",
            sum(sample_results$extraction_time, na.rm
= TRUE) /
            sum(sample_results$n_pages)))
```

Average time per page: 0.96 seconds